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CASE STUDY

MASTER PPC

INDUSTRY	RENEWABLE ENERGY
CUSTOMER	TATA Power Renewable Limited
OBJECTIVE	Master Power Plant Controller (PPC)
LOCATION	Neemuch
PROJECT	TATA Neemuch Plant 1: 160 MW
	TATA Neemuch Plant 2: 170 MW

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CHALLENGES

- Managing control parameters for two plants from a single location
- Controlling each plant individually with slave PPCs
- Generating commands based on the GSS Power Quality Meter reference
- Controlling SVGs in accordance to master PPC commands
- Synchronizing plant events with WRLDC





- Conducting Hardware-in-the-Loop tests
- Performing updated PSSE and PSCAD system studies for the master PPC
- Millisecond resolution of real-time parameters and historical trends
- Optimizing loop time
- CTU and WRLDC standards compliance

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SOLUTION



✔ **Hot Standby PPC Controller:**
- Top Tier Hot Standby PPC Controller
- High processing power
- Highly reliable network configuration

✔ **Optimized Loop Time:**
- Optimized loop time enhancing the input scanning of PQM data
- PPC data processing
- Output writing to slave devices (inverters etc)

✔ **Customized PPC Logic:**
- Common station-based PQM
- Draw a reference from a single point (PQM)
- Distribute commands to the slave plant's PPCs (160 MW and 170 MW)

✔ **Updated System Study Files:**
- Updated PSSE & PSCAD system study files
- Meet simulation output requirements

✔ **Hardware-in-the-Loop Testing:**
- Conducted hardware-in-the-loop tests at CPRI to validate PPC response

✔ **Benchmarking for Compliance:** Benchmarking of PPC system files to comply with CTU & WRLDC

✔ **Event Synchronization Testing:**
- Conduct block-to-WRLDC level tests
- Ensure synchronization of plant events

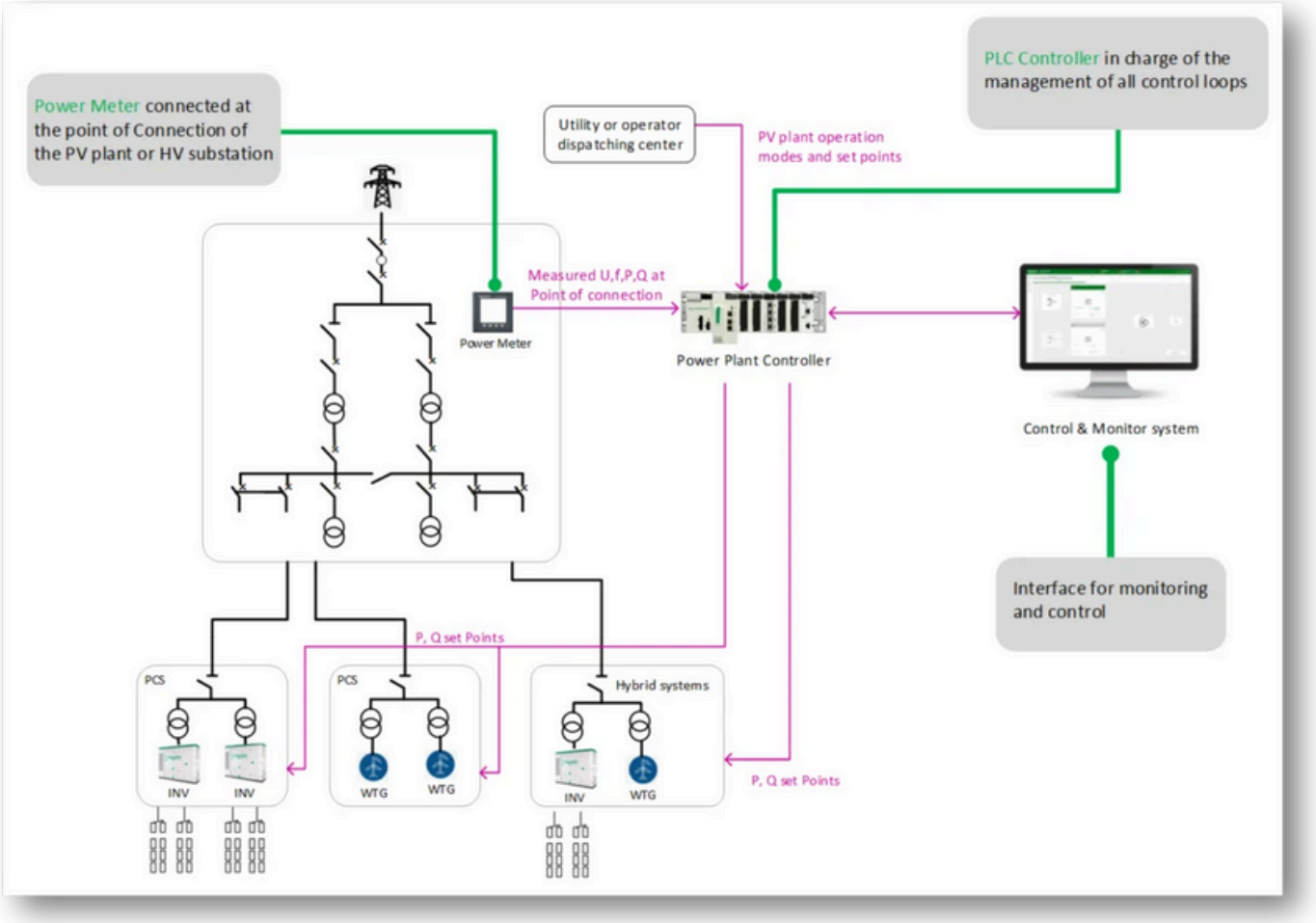
✔ **SVG Integration and Validation:** Successfully integrated, tested, & validated the SVGs to ensure compliance with common POI grid voltage standards

✔ **High-Resolution Data Provision:** Provided PPC operational parameter data with millisecond resolution to WRLDC for plant data analysis



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ARCHITECTURE





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VALUE CREATION

- **Comprehensive Compliance:**
  - Ensure compliance of parameters viz. active power, reactive power, voltage & frequency
  - Single point referencing for integration at both plant and GSS levels
- **Centralized Event Visualization:** Multiple plant events visualised at a single location
- **Fault-Tolerant Network:** Ensures high reliability and availability of operations





- **Unified SVG Operation:** Common SVG operation for multiple plants
- **Transmission Loss Compensation:** Effective compensation for transmission losses
- **Regulatory Compliance:** Ensures compliance to obtain plant charging approvals

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